Completing the odd projective-linear triality geometries

Human Compression project: L023 uniform completion proof

The unrestricted Construction 6 in Leemans–Stokes, arXiv:2208.08215v2, need not produce a geometry, even from a faithful Class III map. This paper proves the missing uniform completion for the odd-characteristic projective-linear family. It therefore supplies actual infinitely many geometries, rather than extrapolating a finite test. The source's advertised family belongs to its authors; this completion proof repairs the missing all-flags step. All programs here are optional checks of the written algebra, not finite premises.

1 A signed-vector intersection bound

Let K be a finite field of odd characteristic. Distinct commuting involutions $a, c \in \mathrm{PSL}_2(K)$ have lifts $i, j \in \mathrm{SL}_2(K)$ with $i^2 = j^2 = -I$ and $ij = -ji$. Indeed a noncentral determinant-one lift of a projective involution is trace zero. If its lift commuted with another such lift, the two trace-zero matrices would be proportional in its two-dimensional matrix centralizer, giving the same projective involution. Thus distinct projectively commuting involutions have anticommuting lifts. Put $k = ij$.

The matrices i, j, k form an orthonormal basis for the trace-zero space W, with bilinear form

$$(X, Y) = -\tfrac{1}{2}\operatorname{tr}(XY), \qquad N(X) = (X, X), \qquad X^2 = -N(X)I.$$

Their multiplication is the quaternion multiplication $ij = k$, $jk = i$, $ki = j$, with reversed products negative. Conjugation gives a faithful action of $\mathrm{PSL}_2(K)$ on W, preserving this form and with determinant one. The kernel consists of scalar matrices. For the determinant statement, conjugation on $M_2(K)$ has determinant $(\det A)^2(\det A^{-1})^2 = 1$; it fixes the scalar summand and preserves the trace-zero summand.

Take a third involution b with lift

$$B = xi + yj + zk, \qquad x^2 + y^2 + z^2 = 1.$$

Suppose b commutes with none of a, c, ac. Then x, y, z are all nonzero. For example $x = 0$ would make B anticommute with i, and hence make b commute projectively with a. Define

$$v_i = (0, -z, y), \qquad v_j = (z, 0, -x), \qquad v_k = (-y, x, 0)$$

in the basis i, j, k. These are the commutators with B, divided by two. Each belongs to $B^\perp$ and is nonzero. Any two are independent: for example $v_i \times v_k = -yB \neq 0$. They consequently span $B^\perp$, which is nondegenerate because $N(B) = 1$. Their pairwise inner products are

$$(v_i, v_j) = -xy, \quad (v_j, v_k) = -yz, \quad (v_k, v_i) = -zx,$$

all nonzero. Individual v_i may be isotropic; we never divide by their norms.

For a norm-one vector B, direct multiplication gives

$$\mathrm{Ad}_b(X) = 2(B, X)B - X. \tag{1}$$

It fixes B and negates $B^\perp$. The involutions a and b therefore both negate v_i; similarly c, b negate v_j, and ac, b negate v_k. Thus

$$H_i = \langle a, b\rangle, \quad H_j = \langle c, b\rangle, \quad H_k = \langle ac, b\rangle$$

respectively stabilize the signed vectors $\{v_i, -v_i\}$, $\{v_j, -v_j\}$, $\{v_k, -v_k\}$.

Lemma 1. *In the adjoint group, the simultaneous stabilizer of any two of these signed vectors is exactly* $\{1, b\}$. *Consequently* $H_i \cap H_j = H_i \cap H_k = H_j \cap H_k = \langle b \rangle$.

Proof. An isometry sending two of the vectors to signed copies must use the same sign on both, since their inner product is nonzero. It is therefore $+1$ or -1 on all of $B^{\perp}$. It preserves the orthogonal line KB; its determinant one forces its action on that line to be $+1$. Thus it is the identity or (1). Faithfulness of the adjoint action identifies these transformations with $1, b$. Both lie in each displayed parabolic, proving the intersections. □

Lemma 2. *There are at most three signed vectors* $\{v, -v\}$ *satisfying*

$$N(v) = N(v_j), \quad (v, v_i) = \pm(v_j, v_i), \quad (v, v_k) = \pm(v_j, v_k).$$

Proof. Choose the unique sign of v making $(v, v_i) = (v_j, v_i)$; the latter is nonzero. If also $(v, v_k) = (v_j, v_k)$, the two independent linear equations give $v = v_j + tB$. Because $(v_j, B) = 0$ and $N(B) = 1$, equality of norms forces $t^2 = 0$, and hence $v = v_j$. For the other sign of the second inner product, the solutions of the two linear equations form an affine line parallel to B. Restricting the norm equation to this line gives a polynomial in t of degree two with leading coefficient one. There are at most two further solutions. This argument also applies when $N(v_j) = 0$. □

2 All flags complete, with the exact residue

Put $G = \langle a, b, c \rangle$, $V = \langle a, c \rangle$, and use the four subgroups H_i, V, H_j, H_k for faces, edges, vertices and Petrie objects. The intersection of H_i with V is $\langle a \rangle$, and the other two intersections are $\langle c \rangle$ and $\langle ac \rangle$. For example $\mathrm{Ad}_c(v_i) = (0, -z, -y)$ is neither v_i nor $-v_i$, because $y, z \neq 0$; the same applies to ac, and the other cases follow cyclically. In particular the three nonedge parabolics are proper.

The rank-three truncation is a faithful thin surface map. At each edge its two faces and two vertices give four flags, a free orbit of V. Every incident face–vertex pair extends using an actual element of the coset intersection; Lemma 1 shows it has exactly two edges. Thus the flags are a single regular G-orbit, and all panels are thin. The flag-adjacency graph is connected by generation. Each vertex or face residue is the cycle of its marked dihedral parabolic. Gluing the corresponding flag triangles therefore gives a closed surface with a two-cell embedded graph. Conversely, unique flag adjacencies imply that a map automorphism fixing a flag fixes all flags. Its full type-preserving incidence group is exactly G.

Theorem 3. *For any such odd-characteristic marking, the four-type coset pregeometry is an incidence geometry. The residue of an incident face–Petrie pair is precisely a path with three vertices and two edges. There are two free* G*-orbits of chambers, and the geometry is not firm.*

Proof. It suffices to consider the standard pair H_i, H_k, since an incident pair of cosets has a common element and can be translated to this pair. A vertex gH_j incident with both has a well-defined signed vector $\{\mathrm{Ad}_g(v_j), -\mathrm{Ad}_g(v_j)\}$. Choose a representative of this coset in H_i. Lemma 1 proves that this map is injective *on vertices incident with this face*: if two representatives $h, h' \in H_i$ have the same signed vector, then $h'^{-1}h$ belongs to

$$H_i \cap \mathrm{Stab}(\{v_j, -v_j\}) = \{1, b\} \subseteq H_j.$$

We do not identify H_j with an entire torus normalizer.

Representing the vertex also by an element of H_k, and using the signed stabilization of v_i, v_k, gives exactly the equations of Lemma 2. There are therefore at most three vertices incident with both H_i, H_k. Three are always present:

$$H_j, \quad aH_j, \quad baH_j. \tag{2}$$

For example $ac \in aH_j \cap H_k$ and $bac \in baH_j \cap H_k$. They are distinct. The first is different from the other two by $H_i \cap H_j = \langle b \rangle$. Equality of the last two would imply $aba \in H_i \cap H_j = \langle b \rangle$, and hence $aba = b$; this would make a, b commute, contrary to hypothesis.

The only edges incident with H_i, H_k are V, bV. To see completeness, choose representatives $h \in H_i$, $l \in H_k$ for any such edge. Their quotient lies in $V = \langle a \rangle \langle ac \rangle$. Absorbing its two factors into h, l gives a representative in $H_i \cap H_k = \{1, b\}$. The vertices of V are H_j, aH_j; those of bV are H_j, baH_j. Thus the exact residue is the asserted path, and every face–vertex–Petrie flag extends.

A flag containing an edge extends because its three nonedge types form the local complete three-partite structure with two choices per type. Any incident pair avoiding edges extends by choosing a common group element. The only additional flags were the triples just handled. Hence every flag extends. The eight local chambers at an edge split into two free V-orbits, yielding two free global G-orbits. At an endpoint of the displayed path the corresponding three-type flag has just one completing edge, so the geometry is not firm. □

3 The required infinite Class III family

For every odd prime power q, the separately reviewed cubic generation theorem supplies a marking of $G = \mathrm{PSL}_2(q^3)$ of Wilson class III. Its triality cycles a, c, ac while fixing b, so the three periods are equal. They exceed two: period two would make all three generators commute, contradicting this group, and period one is incompatible with the triality. Thus b commutes with none of the three endpoint involutions, as required above. Theorem 3 now gives a genuine incidence geometry. The separately proved faithful-map symmetry theorem applies: the edge type has two neighbors in each other type, whereas nonedge objects have $m > 2$ edges, so every correlation fixes the edge type. Restriction to the faithful original map excludes type transpositions. The exact type-permutation quotient is C_3.

Corollary 4. *These are infinitely many pairwise nonisomorphic rank-four incidence geometries with trialities and no dualities. Their full automorphism group is G, their correlation group is $G \rtimes C_3$, and their full automorphism group has exactly two chamber orbits.*

Proof. Every type-preserving correlation restricts to an automorphism of the original map, hence to an element of G. Remove that element; the residual correlation fixes all vertices, edges and faces and can move only Petrie objects. Conjugate it by the known triality to obtain one which moves only vertices and fixes all edges and faces. Its restriction to the original map is an element of G fixing every edge. That edge action is faithful: its kernel is normal and contained in V, whereas the standard simple group $\mathrm{PSL}_2(q^3)$ has no such nontrivial normal subgroup. The residual correlation is therefore the identity. This proves the full automorphism group assertion. The actual order-three triality supplies the complement to G in the correlation group.

The number of edges is $|\mathrm{PSL}_2(q^3)|/4$, which is unbounded and strictly increasing with q. Even an isomorphism allowed to permute types must preserve the intrinsically distinguished edge type. Hence distinct odd q give nonisomorphic geometries. The two-orbit assertion is Theorem 3 with the now identified full automorphism group. □

Dependencies and exact source scope. The entire four-page cubic-generation proof, its stated classification and descent inputs, and the five-page faithful-map symmetry proof are paid dependencies. Simplicity of $\mathrm{PSL}_2(Q)$ for $Q \geq 27$ is retained for the full-group corollary. Completion is proved by the written algebra, with no finite premise. This recovers the odd infinite family, not the false unrestricted Lemma 7. The final broad question about infinite *flag-transitive* families is answered by the separately accepted later Leemans–Stokes–Tranchida work (L015), not by this completion paper. That result is not a premise here. No new question, Lean or numerical credit is claimed.